

Adaptive Multi-Scale Window Segmentation and Attention Interaction Transformer for Intrusion Detection in Industrial Internet of Things

Zhi Yu , Jiaming Pei , and Mingpeng Zhu 

Abstract—As the core of modern industrial automation, Industrial Internet of Things (IIoT) can achieve data collection, control, and remote operations. However, the complex continuous connectivity exposes IIoT to increasing cyber intrusions. Existing intrusion detection methods often fail to adapt to diverse temporal attack patterns, lack effective feature modeling, and perform poorly under class imbalance. To this end, this article proposes a novel transformer-based IIoT intrusion detection method. First, an adaptive multi-scale window segmentation (AMSWS) strategy is proposed to dynamically segment traffic sequences via traffic variation. This can capture short-term and long-term intrusion behaviors. Second, an attention interaction transformer (AITrans) is proposed to enhance feature modeling by enabling inter-head and inter-layer attention interaction. Finally, an adaptive focal loss (AFL) is proposed to mitigate class imbalance by incorporating adaptive class weight and hardness-aware adjustment. This can improve detection performance on minority-class intrusions. Extensive experiments validate the effectiveness of the proposed method for IIoT intrusion detection.

Index Terms—Attention interaction transformer (AITrans), industrial Internet of Things (IIoT), intrusion detection, window segmentation.

I. INTRODUCTION

THE Industrial Internet of Things (IIoT) is revolutionizing industrial automation and related domains by integrating intelligent sensing, data communication, and efficient computation [1], [2], [3], [4]. As IIoT relies on persistent network connectivity, this makes it an attractive target for cyber attackers. Traditional security mechanisms often fall short in complex intrusion environments [5], [6], [7], [8]. Therefore, constructing a robust intrusion detection framework is critical for IIoT security.

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Early methods rely on convolutional neural networks (CNNs) [9], [10], recurrent neural networks (RNNs) [11], [12], long short-term memory networks (LSTMs) [13], [14], and related variants. However, CNNs cannot capture temporal dependencies, while RNNs and LSTMs face vanishing gradients that hinder long-range dependence learning.

Recently, transformers have gained attention with strong modeling abilities in IIoT intrusion detection [15], [16], [17]. However, several limitations remain. First, most methods rely on fixed-size windows, thereby limiting the capture of short-term burst anomalies and long-term stealthy anomalies. Second, most methods lack explicit cross-head and cross-layer information interaction, thereby weakening discriminative representations. Third, most methods struggle to effectively learn features from rare samples, thereby causing degraded performance on low-frequency intrusions.

To this end, this article proposes a novel transformer-based IIoT intrusion detection method.

First, an adaptive multi-scale window segmentation (AMSWS) strategy is proposed to capture heterogeneous temporal patterns for IIoT intrusions. Specifically, when traffic exhibits short-term drastic changes, AMSWS applies smaller windows to capture burst threats. Conversely, when traffic exhibits long-term hidden threats, AMSWS expands windows to analyze longer sequential features. This overcomes the rigidity of fixed-window designs by adjusting windows via traffic variation. Although some adaptive window techniques have been explored in time-series analysis [18], [19], [20], [21], [22], most methods generally involve heavy algorithmic structures, intricate parameter tuning, and high computational overhead. In contrast, AMSWS adopts a lightweight and continuously adjustable mechanism via traffic variation. Without relying on complex statistical analysis, AMSWS generates non-overlapping multi-scale windows, which are directly integrated with subsequent feature mappings to effectively capture both short-term burst anomalies and long-term stealthy anomalies.

Then, an attention interaction transformer (AITrans) is proposed to enhance traffic modeling. While recent hierarchical transformer variants for intrusion detection have improved feature extraction [23], [24], [25], [26], [27], most architectures compute multiple attention heads independently within the same layer, lacking explicit inter-head attention interaction. This hinders the learning of complementary patterns across different subspaces. In addition, few existing models incorporate effective

TABLE I
COMPARISON OF REVIEWED IOT INTRUSION DETECTION METHODS

Reference	Methodology	Highlights	Limitations
Qathrady et al. [28]	CNN + Attention	Combine CNN and self-attention mechanism to extract the most significant traffic features	Cannot capture temporal features, cannot effectively address class imbalance
Li et al. [29]	CNN + RNN + SVM	Combine CNN and RNN simultaneously to extract spatial features and temporal features	Cannot capture long-term dependencies, cannot effectively address class imbalance
Zeng et al. [30]	VAE + GAN + PSO	Utilize VAE to generate multivariate series, utilize PSO to optimize parameters and neural architectures	Cause high computational cost from PSO and adversarial training, cause potential noise from generated data
Lu et al. [31]	CNN + NAS + HPO	Combine NAS and HPO to design the lightweight CNN, utilize the hybrid variable-length encoding to characterize neural architectures and parameters	Cause high computational cost from NAS and HPO, cannot capture temporal features, cannot effectively address class imbalance
Vadakkethil et al. [32]	BiLSTM + MOA	Utilize MOA to select important features, utilize BiLSTM to capture bidirectional temporal dependencies	Cause high computational cost from MOA, cannot effectively address class imbalance
Yang et al. [33]	BiGRU + CNN + Attention	Combine BiGRU, CNN, and attention to improve feature representation, utilize a hybrid sampling strategy to address the bias learning caused by data imbalance	Cause high model complexity and computational cost through multi-module network fusion and hybrid sampling strategy
Liang et al. [25]	Transformer + DWT	Combine DWT and transformer to extract frequency information of traffic data	Cannot capture complex temporal characteristics, cannot effectively address class imbalance
Fares et al. [34]	Transformer + KAN	Replace MLP with KAN to reduce parameters, enhance transformer to improve feature representation	Cause high training cost for KAN function optimization, cannot effectively address class imbalance
Moghaddam et al. [35]	Transformer + GAN	Combine transformer for feature modeling with GAN for data generation to enhance feature diversity	Cause high computational cost from adversarial training, cause potential noise from generated data
Wang et al. [36]	Transformer + DM	Utilize DM to generate balanced data, utilize time-series transformer to capture local and global features	Cause high training cost from iterative diffusion processes, cause potential noise from generated data
Zeng et al. [37]	Transformer + Causal Attention	Utilize causal attention to decouple spurious features, utilize causal transformer to learn unknown attacks	Cause high computational cost from complex causal analysis, cannot effectively address class imbalance
Wang et al. [38]	Transformer + GADF	Transform traffic into images to enhance features, utilize focal loss to address class imbalance	Cause high computational cost from image generation, rely on complex imaging/parameter tuning

mechanisms for inter-layer information reuse. This limits the hierarchical representation ability. AITrans introduces inter-head interaction attention (IHIA) and inter-layer interaction attention (ILIA) to solve these issues. The former can facilitate inter-head attention interaction to share information across different attention subspaces. The latter can facilitate inter-layer attention interaction to guide higher transformer layers referencing information in lower transformer layers.

Finally, an adaptive focal loss (AFL) is proposed to mitigate class imbalance in IIoT intrusion detection. Different from embedding complex generative models to synthesize rare samples, AFL incorporates adaptive class weight (ACW) and hardness-aware adjustment (HAA) in focal loss. The former can dynamically adjust the class weight based on the class frequency, which ensures that the model is attentive to minority-class rare anomalies. The latter can dynamically adjust the focusing parameter based on the sample confidence, which assigns higher loss values to low-confidence hard anomalies.

The contributions of this article are summarized as follows.

- 1) An AMSWS strategy is proposed to dynamically adjust windows. Unlike prior fixed-size windowing methods that lack adaptability to diverse intrusion patterns or complex adaptive windowing methods that incur high computational overhead, AMSWS employs a lightweight window adjustment strategy via traffic variation. AMSWS utilizes smaller windows to capture short-term burst anomalies and larger windows to capture long-term stealthy anomalies. This enables the model to adaptively align with transient and persistent intrusion patterns.
- 2) An AITrans is proposed to enhance traffic modeling. Unlike existing transformers that lack explicit cross-head and cross-layer information exchange, AITrans utilizes IHIA and ILIA to improve traffic representation. AITrans enables the extraction of complementary information across attention subspaces and allows higher layers to effectively utilize the cues learned in lower layers.
- 3) An AFL is proposed to mitigate class imbalance. Unlike embedding complex generative models to synthesize

rare samples, AFL offers a more efficient alternative by directly optimizing the learning with a tailored loss. It integrates ACW and HAA to tune the class weight and the focusing parameter. AFL can guide AITrans to focus on rare samples.

II. RELATED WORKS

Existing IoT intrusion detection methods can be divided into non-transformer methods and transformer methods. Table I summarizes highlights and limitations of related methods.

A. Intrusion Detection via Non-Transformer Methods

Qathrady et al. [28] proposed a self-attention CNN termed SACNN to isolate the most relevant traffic features for intrusion detection. It incorporates the self-attention mechanism to compute attention weights from the input data, followed by the convolutional layers that utilize these weighted features to perform accurate detection.

Li et al. [29] proposed an anomaly-based intrusion detection method termed CRSF. It uses the dimension transformation to convert input data into 2-D images. Spatial features are extracted via the CNN, and the generated features are processed via the RNN to capture temporal dependencies. A support vector machine (SVM) is employed as the classifier to project the features and learn nonlinear decision boundaries.

Zeng et al. [30] proposed an unsupervised intrusion detection method termed EvoAAE. Specifically, a variational autoencoder (VAE) based on the generative adversarial network (GAN) is designed to adversarially generate multivariate time series. Then, the particle swarm optimization (PSO) with an efficient binary encoding is designed to optimize hyper-parameters and neural architectures in the adversarial VAE.

Lu et al. [31] proposed an intrusion detection method termed MODEO-CNN based on the multi-objective joint optimization, which integrates the neural architecture search (NAS) and the hyper-parameter optimization (HPO). It introduces a hybrid variable-length encoding combining binary and integer

encoding to represent the number of blocks, network topology, block parameters, and key hyper-parameters.

Vadakkethil et al. [32] proposed an intrusion detection method that combines the Mayfly optimization algorithm (MOA) with the bidirectional LSTM (BiLSTM). MOA is used for feature selection by balancing exploration and exploitation to identify the most relevant features. BiLSTM is used for processing input to capture long-range temporal dependencies.

Yang et al. [33] proposed an intrusion detection method that integrates the attention, the bidirectional gated recurrent units (BiGRU), and the Inception CNN. A mixed sampling strategy is adopted to mitigate the impact of data imbalance, while denoising techniques are applied to reduce noise introduced during resampling.

Overall, intrusion detection methods via non-transformer perform effectively in learning local spatial features and modeling short-term temporal dependencies. Besides, these methods exhibit clear advantages in terms of lightweight computation and relatively low model complexity. However, the limited receptive fields and strictly sequential computation hinder effective modeling of long-range temporal dependencies. Furthermore, most methods rely on complex sampling strategies or generative algorithms to mitigate class imbalance, which increases design complexity.

B. Intrusion Detection via Transformer Methods

Liang et al. [25] proposed a multi-level intrusion detection model that integrates the transformer with the discrete wavelet transform (DWT). By embedding DWT and multi-wavelet learning into the transformer, the model effectively captures frequency-based features from IoT traffic, improving the detection accuracy for intrusion behaviors.

Fares et al. [34] proposed an intrusion detection model termed TFKAN, which incorporates the Kolmogorov-Arnold network (KAN) layers in place of the multi-layer perceptron (MLP) layers. The KAN utilizes learnable univariate functions instead of fixed weights, allowing for more compact and expressive representations. Therefore, TFKAN achieves competitive performance with fewer parameters.

Moghaddam et al. [35] proposed a hybrid architecture that integrates the transformer for capturing temporal and contextual dependencies with the GAN for generating synthetic data to address class imbalance. Besides, an improved non-dominated sorting biogeography-based optimization algorithm is introduced to optimize hyper-parameters.

Wang et al. [36] proposed an intrusion detection method termed DIFT to address class imbalance by combining the diffusion model (DM) with the transformer. The former learns features of minority-class samples and generates more balanced data. To improve processing efficiency, a patching method is applied to shorten the input and enhance local features. Subsequently, a time transformer is designed to extract local and global features.

Zeng et al. [37] proposed a causal transformer for intrusion detection. The first module classifies samples into known categories using prior knowledge. The second module employs

causal attention to learn causal features. The final module groups samples differing from known categories as unknown types and classifies these samples with a causal transformer.

Wang et al. [38] proposed an intrusion detection framework termed ImagTIDS, which integrates the Gramian angular difference fields (GADF) with the transformer. The traffic is transformed into images using GADF to retain temporal and global features. ImagTrans then extracts local and global features. By leveraging self-attention, the model dynamically adjusts attention weights to emphasize critical features while reducing the impact of redundant information.

Overall, intrusion detection methods via transformer demonstrate significant strengths in capturing long-range dependencies to improve robustness. Besides, transformer variants can integrate other information to enrich features. However, these methods face some limitations. First, these methods mainly rely on fixed-size windows, which fail to adapt to the diverse temporal patterns of intrusions. This limits the ability to capture short-term burst anomalies and long-term stealthy anomalies. Although some adaptive windowing strategies have been explored in time-series analysis, most methods involve complex algorithms and analysis. Then, these transformer-based methods compute attention heads independently within the same layer. The explicit inter-head and inter-layer information interaction are ignored. This impedes cross-subspace learning and weakens hierarchical representation. Finally, these methods cannot effectively address class imbalance. Although some methods utilize generative models to synthesize minority-class samples, such methods incur high computational cost and introduce synthetic noise.

To solve the above three issues, AMSWS, AITrans, and AFL are proposed. AMSWS embeds a lightweight window adjustment strategy to adaptively divide traffic for capturing short-term burst anomalies and long-term stealthy anomalies. AITrans embeds IHIA and ILIA to facilitate inter-head and inter-layer attention interaction for improving traffic modeling. AFL embeds ACW and HAA to tune the class weight and the focusing parameter for mitigating class imbalance.

III. METHOD

A. Overall Architecture

The proposed architecture is shown in Fig. 1. AMSWS adjusts windows via traffic variation. AITrans enhances traffic modeling via attention interaction. AFL mitigates class imbalance by tuning the class weight and the focusing parameter.

B. Adaptive Multi-Scale Window Segmentation

Considering the limitations of fixed-window strategies, an AMSWS is proposed to dynamically adjust windows. It consists of traffic preprocessing, traffic variation analysis, adaptive window adjustment, and traffic segmentation.

1) *Traffic Preprocessing*: The input is a complete segment of network traffic including T time steps as follows:

$$X = \{x_1, x_2, \dots, x_t, \dots, x_T\} \quad (1)$$

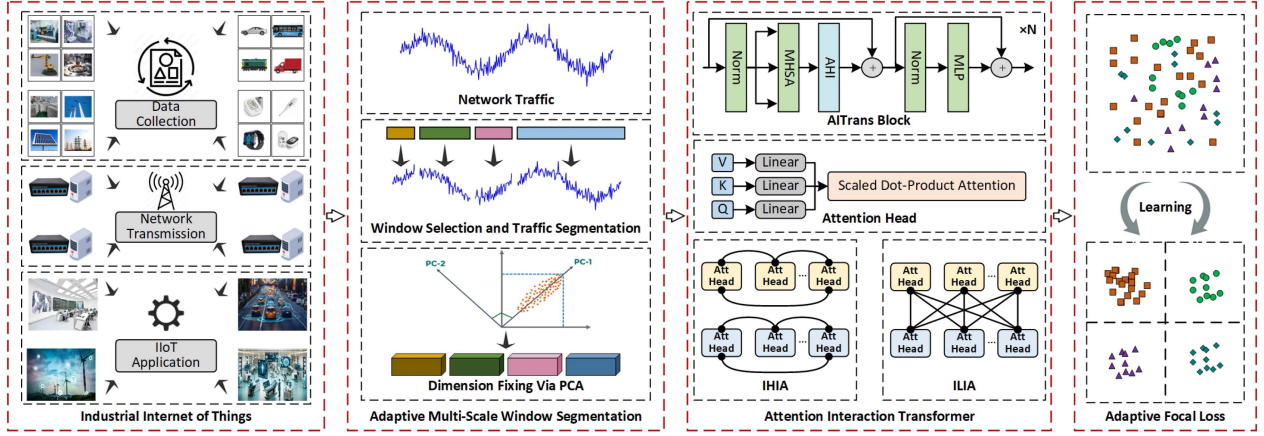


Fig. 1. Overall architecture of our proposed method for IIoT intrusion detection. It consists of AMSWS, AITrans, and AFL. First, AMSWS dynamically segments traffic with adaptive windows. It effectively captures anomalous traffic patterns at different temporal scales. The segmented traffic is mapped to a fixed dimension with PCA. Next, the traffic sequence is fed into AITrans for feature modeling. The multi-head self-attention is enhanced through AHI. It integrates IHIA and ILIA to improve feature representation. Finally, AFL integrates ACW and HAA to tune the class weight and the focusing parameter. It can guide AITrans to focus on rare samples and learn more discriminative features.

where x_t is the traffic data at the t th time step.

Then, X is normalized with z-score as follows:

$$\tilde{x}_t = \frac{x_t - \mu}{\sigma} \quad (2)$$

$$\tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_T\}. \quad (3)$$

2) Traffic Variation Analysis: For normalized traffic, the first-order difference analysis is first performed to measure the traffic change between adjacent times as follows:

$$\Delta x_t = \tilde{x}_t - \tilde{x}_{t-1}. \quad (4)$$

Considering that the change in a single time step may be affected by noise, the mean variation rate within the sliding window is calculated to measure local fluctuations as follows:

$$R_t = \frac{1}{M} \sum_{i=t-M+1}^t |\Delta x_i| \quad (5)$$

where R_t is the traffic variation rate at time step t , M is the sliding window with a fixed size to capture the traffic change.

3) Adaptive Window Adjustment: The objective of AMSWS is to adjust windows via traffic variation. This allows the system to adapt to different temporal scales of anomalous traffic patterns. First, the window adjustment factor is calculated to control the range of window variations as follows:

$$\alpha_g = \frac{1}{1 + e^{-k(R_t - \theta)}} \quad (6)$$

where k and θ jointly determine the window sensitivity to traffic variation, thereby controlling the adaptive window adjustment. Specifically, k regulates the response intensity of the window to traffic variation magnitude. θ controls the sensitivity threshold to the traffic variation rate. As R_t approaches the threshold, the variation in α_g becomes more significant. By changing k and θ , the corresponding adjustments can be made to the window size.

Then, based on the window adjustment factor, the adaptive window size is calculated as follows:

$$W_g = W_{\max} - (W_{\max} - W_{\min}) \cdot \alpha_g \quad (7)$$

where $W_{\min}$ and $W_{\max}$ are min and max windows.

α_g is negatively correlated with the window size. Specifically, as R_t gradually increases, α_g also gradually increases and approaches 1, while the corresponding window gradually decreases to $W_{\min}$. At this point, a smaller window can ensure finer segmentation of short-term anomalies. On the contrary, as R_t gradually decreases, α_g also gradually decreases and approaches 0, while the corresponding window gradually increases to $W_{\max}$. At this point, a larger window can facilitate the modeling of long-term anomalies.

Finally, through window adjustments, G non-overlapping windows are obtained. The multi-scale window set is as follows:

$$\mathcal{W} = \{W_1, W_2, \dots, W_G\}. \quad (8)$$

4) Traffic Segmentation: Based on the multi-scale window set, the normalized traffic sequence $\tilde{X}$ is segmented accordingly. For each window, the traffic segment is defined as follows:

$$F_g = \{\tilde{x}_{\text{start}_g}, \tilde{x}_{\text{start}_g+1}, \dots, \tilde{x}_{\text{start}_g+W_g-1}\}. \quad (9)$$

The adjustment of window lengths ensures the model adapts to various types of anomalies. Finally, after window segmentation, G traffic segments are obtained as follows:

$$\mathcal{F} = \{F_1, F_2, \dots, F_G\}. \quad (10)$$

Since the lengths of different traffic segments vary and the subsequent network requires input data to keep the same dimensions, all traffic segments are transformed into a uniform shape via principal component analysis (PCA) and linear mapping as follows:

$$F'_g = f_{\text{LM}}(f_{\text{PCA}}(F_g)) \quad (11)$$

where f_{LM} is linear mapping and f_{PCA} is PCA processing.

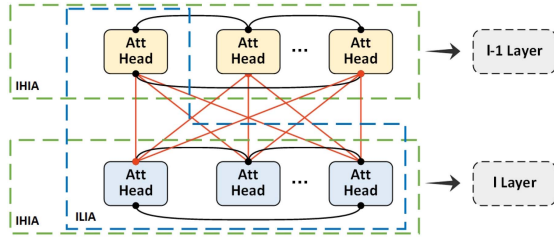


Fig. 2. AHI module. It consists of IHIA and ILIA.

Finally, the generated traffic segment set is as follows:

$$\mathcal{F}' = \{F'_1, F'_2, \dots, F'_G\}. \quad (12)$$

C. Attention Interaction Transformer

Recently, transformers have achieved significant progress in traffic modeling and intrusion detection. However, these methods have some limitations. First, information sharing among attention heads is lacking. Second, inter-layer information flow is limited. Therefore, this article proposes an AITrans. It introduces an attention-head interaction (AHI) mechanism as illustrated in Fig. 2 to enhance information interaction. AHI consists of IHIA and ILIA.

1) *Inter-head Interaction Attention*: In transformers, multiple attention heads focus on different feature patterns independently. Each head computes attention distributions in isolation, and information sharing is not efficient. This leads to situations where some attention heads may redundantly focus on some specific traffic features, while others may entirely overlook these critical patterns. Therefore, this article proposes an IHIA mechanism. It enables different attention heads to share information and enhance feature representation capability. First, the attention from all heads is aggregated to integrate multi-head information as follows:

$$A_f^{(l)} = \sum_{h=1}^H a_h A_h^{(l)} \quad (13)$$

where $A_h^{(l)}$ is the attention calculated by the h th attention head in the l th layer, a_h is the weight for this head and satisfies

$$a_h = \frac{\exp(\lambda_h)}{\sum_{j=1}^H \exp(\lambda_j)} \quad (14)$$

where λ_h is a trainable parameter.

Then, utilize the aggregated attention information to update each attention head as follows:

$$A_h^{(l)} = \alpha A_h^{(l)} + (1 - \alpha) A_f^{(l)}, \quad h = 1, 2, \dots, H \quad (15)$$

where α controls fusion of original and aggregated attention.

IHIA facilitates cross-attention information exchange between attention heads. It avoids the information fragmentation caused by the independent computation of attention heads and improves global modeling capabilities.

2) *Inter-layer Interaction Attention*: In transformers, the multi-head self-attention computation in each layer relies solely on the input from the current layer, and attention information across different layers cannot interact directly. To overcome this

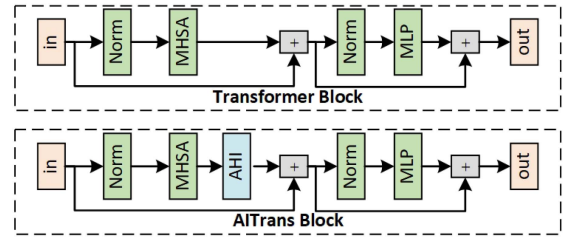


Fig. 3. Transformer block and AITrans block.

issue, this article proposes an ILIA mechanism. This enables the transformer to share attention information across layers, thereby enhancing overall modeling capability. To establish inter-layer information flow, residual connections are introduced, allowing each current-layer attention to reference the previous adjacent layer as follows:

$$B_h^{(l)} = B_h^{(l)} + \sum_{j=1}^H b_j B_j^{(l-1)} \quad (16)$$

where $B_h^{(l)}$ is the h th attention head in the current layer, $B_j^{(l-1)}$ is the j th attention head in the previous adjacent layer, b_j is the weight and satisfies

$$b_j = \frac{\exp(\mu_j)}{\sum_{m=1}^H \exp(\mu_m)} \quad (17)$$

where μ_j is a trainable parameter.

With residual connections, each layer can maintain its own information while incorporating historical information from the previous adjacent layer. It enhances the ability to capture anomalous patterns. In addition to introducing all attention heads from the previous adjacent layer, this article further computes an aggregated representation of all lower-layer attention heads to strengthen the information transmission as follows:

$$B_a^{(l)} = \sum_{s=1}^{l-1} \sum_{j=1}^H B_j^{(s)}. \quad (18)$$

Then, perform a weighted fusion of the inter-layer aggregated attention with the current-layer attention as follows:

$$B_h^{(l)} = \beta B_h^{(l)} + (1 - \beta) B_a^{(l)} \quad (19)$$

where β controls fusion of different types of attention.

ILIA enhances inter-layer information flow and improves inter-layer information sharing. Additionally, ILIA enables higher-layer features to retain lower-layer details. It allows local features extracted at lower layers to influence global feature modeling at higher layers, thereby improving the ability to capture complex traffic patterns.

Finally, by integrating IHIA and ILIA, AHI produces the processed attention as follows:

$$C_h^{(l)} = A_h^{(l)} + B_h^{(l)}. \quad (20)$$

As shown in Fig. 3, AHI is combined with the traditional transformer block to construct an AITrans block. AITrans blocks are stacked to construct AITrans for traffic modeling.

D. Adaptive Focal Loss

In IIoT intrusion detection, anomalous traffic is only triggered by attacks or abnormal events. The class imbalance causes the model to favor learning normal traffic patterns, thereby reducing detection performance on minority classes. Therefore, focal loss [39] is introduced to enhance the detection capability for minority-class samples. However, this loss still suffers from the issue of fixed parameters, which makes it difficult to adapt to different data distributions.

Therefore, AFL is proposed to improve focal loss by incorporating ACW and HAA. ACW adjusts the class weight at the global level via class frequency. It aligns with the sample balancing principle and mitigates majority-class dominance. Inspired by hardness-aware learning, HAA modulates the focusing parameter via sample confidence to focus on hard samples. Unlike using complex generative models to generate samples, AFL directly improves the loss with lightweight and adaptive mechanisms. This allows the model to better handle rare anomalies and avoid the computational cost from generative methods. The standard focal loss is as follows:

$$\mathcal{L}_{FL} = -(1 - p_t)^\gamma \log(p_t) \quad (21)$$

where p_t is the sample confidence, γ is the focusing parameter.

To enhance focal loss in handling imbalanced data, AFL introduces ACW and HAA as follows:

$$\mathcal{L}_{AFL} = -\omega_t(1 - p_t)^{\gamma_t} \log(p_t) \quad (22)$$

where ω_t corresponds to ACW, which adjusts the class-specific weight via class frequency to focus on minority-class samples. γ_t corresponds to HAA, which adjusts the contribution of traffic via sample confidence to focus on hard samples.

1) Adaptive Class Weight: In the training, minority-class samples struggle to exert sufficient influence on the model, making it difficult to accurately detect rare anomalous traffic. Therefore, this article introduces ACW to dynamically adjust the weights for different classes. The core idea is to modify the class weight via the corresponding class frequency, thereby enhancing the influence of minority classes as follows:

$$\omega_t = \frac{1}{\sqrt{f_t} + \epsilon} \quad (23)$$

where f_t denotes the frequency of class t , it is computed globally from the entire training set.

When the class frequency f_t is high (majority-class traffic), the weight ω_t is relatively small, preventing the model from being overly biased toward majority-class traffic. Conversely, when the class frequency f_t is low (minority-class traffic), the weight ω_t is larger, enhancing the model's focus on minority-class traffic. This adaptive balance improves the model's ability to detect anomalies in imbalanced datasets. ACW strengthens the impact of minority-class samples, preventing the model from overfitting to majority-class traffic. Besides, ACW achieves dynamic weight assignment according to the class distribution. It adaptively adjusts class importance at the global level. This data-driven adjustment reflects the class distribution during training and provides a theoretically grounded yet lightweight solution to class imbalance.

2) Hardness-Aware Adjustment: Focal loss mitigates the impact of easy samples by assigning higher loss to hard samples. However, it employs a fixed focusing parameter. This rigid focus is difficult to adapt to different data distributions. It has some positive effects on hard samples, but may not be sufficiently fine-grained. Therefore, this article introduces HAA, which dynamically adjusts the focusing parameter based on the sample confidence as follows:

$$\gamma_t = \gamma_{\max} - (\gamma_{\max} - \gamma_{\min})(1 - p_t) \quad (24)$$

where $\gamma_{\min}$ and $\gamma_{\max}$ are min and max focusing parameter, and p_t is the sample confidence. For easy samples (p_t is high), γ_t is close to $\gamma_{\max}$, the loss will decrease. Similarly, for hard samples (p_t is low), γ_t is close to $\gamma_{\min}$, the loss will increase.

By dynamically adjusting γ_t , HAA ensures that the model focuses more on hard samples. This reduces over-reliance on easy samples and enhances the ability to detect minority-class anomalies. HAA uses a lightweight, hardness-driven adaptation mechanism that reflects sample difficulty during training. By dynamically adjusting the focusing parameter via sample confidence, HAA achieves more flexible and data-driven optimization, thereby better highlighting hard samples. This provides a theoretically sound and practically effective solution for fine-grained optimization in class balance.

By integrating ACW and HAA, AFL not only adjusts the class weight via class frequency but also modifies the focusing parameter via sample difficulty. This focuses on minority-class samples and mitigates class imbalance. When combined with AITrans, it further enhances intrusion detection capabilities.

IV. EXPERIMENTS

A. Datasets

This article constructs IIoTA and IIoTB datasets. Each sample is captured using Wireshark and system logging modules. The intrusion categories in IIoTA and IIoTB are aligned with TON_IoT [6], thereby covering the major intrusion types commonly observed in IIoT systems. Specifically, IIoTA and IIoTB contain ten categories: Normal, DoS, DDoS, Scanning, MITM, Injection, XSS, Password, Backdoor, and Ransomware.

In real-world IIoT networks, normal traffic typically dominates, while anomalous traffic accounts for a relatively small proportion. Moreover, the frequency of occurrence varies across different intrusions. To reflect this realistic setting, different numbers of samples are collected for different categories. IIoTA contains 218 863 samples, the quantities for each category are 100 392, 12 032, 10 679, 9904, 13 561, 8996, 9104, 11 019, 9531, and 33 645. IIoTB contains 436 794 samples, the quantities for each category are 231 123, 31 529, 23 638, 19 177, 23 498, 22 916, 19 973, 36 145, 20 561, and 8234. These distributions are intended to realistically represent traffic imbalances.

Each traffic sample is represented as a fixed-length time series with 300 time steps. This setting can capture significant intrusion patterns and retain sufficient contextual information. It also avoids gradient vanishing and excessive computational overhead caused by overly long sequences. In addition, many studies on intrusion detection adopt sequence lengths of around 300 time steps for input construction [17], [40].

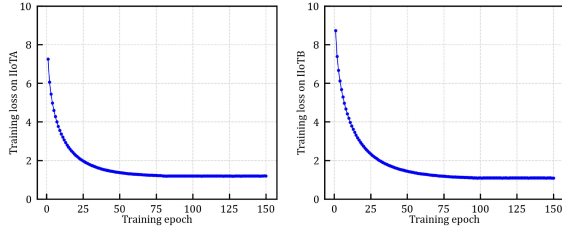


Fig. 4. Analysis of training process.

Each time step contains a 128-D feature. This setting provides sufficient feature space to represent complex traffic patterns. This also avoids the excessive computational cost and overfitting risks that arise from higher dimensions. Besides, many studies on intrusion detection adopt similar feature dimensions [10], [41].

For IIoTA/IIoTB, 70% samples are treated as the training set, 30% samples are treated as the test set. This split is widely adopted in intrusion detection studies [5], [42]. The 70% training portion provides enough data to learn diverse traffic patterns and intrusion behaviors, while the 30% test portion ensures robust evaluation of generalization performance.

B. Implementation Details

All experiments are conducted on a workstation equipped with NVIDIA RTX 3090 GPU, Intel Core i9-12900 K CPU, Ubuntu 20.04, Python 3.7, and PyTorch 1.8.1. Accuracy (Acc), and F1 score are evaluation metrics.

The Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ is employed to train AITrans. These are default settings in the Adam paper and have been validated in deep learning [43]. The initial learning rate is 0.001, which allows the model to converge efficiently while maintaining stability. A decay factor of 0.1 is set to reduce the learning rate during the training stage, which avoids oscillation and overfitting. These settings are commonly adopted in intrusion detection studies [44], [45].

The common batch sizes are 32, 64, 128, and 256 in intrusion detection. In a reasonable range, larger batches aid training stability and convergence speed. However, excessively large batches cause overfitting and exceed GPU memory. Therefore, a batch size of 128 is adopted to strike a balance between training speed and generalization. Each batch includes eight categories, each category with 16 samples. These settings are commonly adopted in intrusion detection studies [46], [47].

The parameters in (7) are $W_{\min} = 10$ and $W_{\max} = 100$. Setting $W_{\min} = 10$ captures short-term fluctuations to detect rapidly changing and bursty anomalies. Setting $W_{\max} = 100$ covers longer intrusion patterns to detect slow and stealthy anomalies. If the window is too small, contextual information may be lost. If the window is too large, critical features may be smoothed out. Therefore, setting $W_{\min} = 10$ and $W_{\max} = 100$ can provide a balance that enables the model to effectively capture short-term and long-term intrusions.

The parameters in (24) are $\gamma_{\min} = 1$ and $\gamma_{\max} = 5$. The original focal loss paper recommends $\gamma = 2$. Building on this, the range of γ is further extended to allow adaptation to samples of varying difficulty. If γ is too small, the loss may degenerate

into cross-entropy loss. It fails to suppress the dominance of easy samples and address class imbalance. If γ is too large, the loss contributions of samples are excessively amplified, causing gradient fluctuations and unstable training. Therefore, setting $\gamma_{\min} = 1$ and $\gamma_{\max} = 5$ can effectively guide the model to suppress easy samples and to focus on hard samples for mitigating the issue of class imbalance.

The other parameters are: $M = 40$ in (5), $k = 3$ and $\theta = 0.6$ in (6), $\alpha = 0.3$ in (15), $\beta = 0.6$ in (19). These parameters are determined through comparative experiments in Figs. 5–10.

C. Comparison With State-of-the-Art Methods

To validate the feasibility of the proposed method, comparative experiments are conducted in Table II. ACNN-IDS [48], INIDS [49], NID-BAI [33], DINet [50], and VAE-WACGAN [51] are non-transformer methods. DWTrans [25], BBO-CFAT [26], MF-Net [52], ImagTIDS [38], and FlowTransformer [53] are transformer methods. The proposed method can achieve the best intrusion detection performance, with the Acc/F1 score of 99.1%/98.7% on IIoTA and 99.6%/99.2% on IIoTB. Compared to other methods, the proposed approach improves the Acc by 1.0%~3.1% and 1.1%~2.5% on IIoTA and IIoTB. It also improves the F1 score by 1.3%~3.3% and 1.2%~2.8% on two datasets. When compared to the best-performing transformer-based method FlowTransformer, AITrans can achieve the Acc/F1 score improvement of 1.0%/1.3% on IIoTA and 1.1%/1.2% on IIoTB.

The complexity analysis is shown in Table II. To evaluate the model complexity, this article considers two categories of metrics based on the association with complexity. The first category of metrics has a relatively strong correlation to complexity, including Parameters (Params), FLOPs, GPU Usage (GU), CPU Usage (CU), GPU Power (GP), CPU Power (CP), Inference Time (IT) per sample, GPU Energy (GE), and CPU Energy (CE). The lower these metrics, the lower the model complexity. Among them, Params, FLOPs, and IT are three key metrics. Params represents the number of trainable parameters, namely the total amount of weights and biases in the model. FLOPs represents the number of floating-point operations in a single inference, reflecting the arithmetic computations during one forward pass. IT represents the time taken for a sample to be processed from input into the network to final classification. In general, under the same software and hardware conditions, a more complex model tends to exhibit larger Params, higher FLOPs, and longer IT. However, the relationship among Params, FLOPs, and IT is not strictly proportional, because these metrics depend not only on model size but also on architectural design choices and data processing strategies. The second category of metrics has a relatively weak correlation to complexity, including Inference Time/Parameters (IT/Params), GPU Energy/Parameters (GE/Params), and CPU Energy/Parameters (CE/Params). These metrics provide additional insights into algorithmic efficiency per parameter, but they may not directly reflect overall complexity. When Params increases significantly, these metrics may decrease even if IT, GE, and CE are higher. Therefore, a more complex model with larger Params may yield a lower IT/Params, GE/Params, and CE/Params.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS

Method	Backbone	IIoTA		IIoTB		Model Complexity											
		Acc (%)	F1 score (%)	Acc (%)	F1 score (%)	Params (M)	FLOPs (G)	GU (%)	CU (%)	GP (W)	CP (W)	IT (ms)	GE (kWh)	CE (kWh)	IT/Params (ms/M)	GE/Params (kWh/M)	CE/Params (kWh/M)
ACNN-IDS [48]	Non-transformer	96.0	95.6	97.1	96.4	21.7	5.8	15.3	35.9	123	62	3.1	10.6E-8	5.3E-8	0.14	0.49E-8	0.24E-8
INIDS [49]	Non-transformer	96.4	95.9	97.5	96.7	20.3	7.3	17.3	42.7	135	69	5.3	19.9E-8	10.2E-8	0.26	0.98E-8	0.50E-8
NIDS-BAI [33]	Non-transformer	96.9	96.0	97.7	97.1	25.7	9.6	17.2	49.1	139	72	5.5	21.2E-8	11.0E-8	0.21	0.82E-8	0.43E-8
DINet [50]	Non-transformer	96.5	95.4	97.2	96.9	28.5	8.1	11.6	37.2	128	60	4.2	14.9E-8	7.0E-8	0.15	0.52E-8	0.25E-8
VAE-WACGAN [51]	Non-transformer	96.3	96.1	97.2	96.0	4.4	3.7	8.5	29.4	112	56	2.6	8.1E-8	4.0E-8	0.59	1.84E-8	0.91E-8
DWTrans [25]	Transformer	97.2	96.8	97.8	97.1	138.2	23.8	34.1	69.1	179	92	9.6	47.7E-8	24.5E-8	0.07	0.35E-8	0.18E-8
BBO-CFAT [26]	Transformer	97.6	97.1	98.0	97.4	49.6	31.3	22.5	55.4	147	84	14.4	58.8E-8	33.6E-8	0.29	1.19E-8	0.68E-8
MF-Net [52]	Transformer	97.5	97.0	98.2	97.7	74.7	29.6	31.1	62.6	162	86	12.9	58.1E-8	30.8E-8	0.17	0.78E-8	0.41E-8
ImagTIDS [38]	Transformer	97.8	97.3	98.4	97.3	103.7	33.5	39.2	74.9	174	88	23.1	111.7E-8	56.5E-8	0.22	1.08E-8	0.54E-8
FlowTransformer [53]	Transformer	98.1	97.4	98.5	98.0	49.7	72.6	28.5	59.9	155	79	12.2	52.5E-8	26.8E-8	0.25	1.06E-8	0.54E-8
AITrans (w/o IHIA/ILIA)	Transformer	97.2	96.4	98.1	97.3	37.2	18.8	19.1	52.2	147	75	6.6	27.0E-8	13.8E-8	0.18	0.73E-8	0.37E-8
AITrans	Transformer	99.1	98.7	99.6	99.2	37.2	32.6	23.8	61.7	150	78	10.8	45.0E-8	23.4E-8	0.29	1.21E-8	0.63E-8

For model complexity, AITrans has 37.2 M Params, 32.6 G FLOPs, 23.8% GU, 61.7% CU, 150 W GP, 78 W CP, 10.8 ms IT, 45.0E-8 kWh GE, 23.4E-8 kWh CE, 0.29 ms/M IT/Params, 1.21E-8 kWh/M GE/Params, and 0.63E-8 kWh/M CE/Params. Compared to non-transformer intrusion detection methods, AITrans exhibits higher complexity due to the intensive attention computations inside the transformer. Compared to transformer-based methods, AITrans has the smallest Params. FLOPs, GU, CU, GP, CP, GE, and CE are also relatively low, and the IT is relatively fast, with only a 1.2 ms increase compared to DWTrans [25]. Due to the relatively small Params in AITrans, the IT/Params, GE/Params, and CE/Params show certain increases compared to most transformer-based methods. Among all methods, VAE-WACGAN [51] exhibits relatively low complexity, achieves the fastest inference speed, and attains the lowest energy consumption. It obtains the lowest Params, FLOPs, GU, CU, GP, CP, IT, GE, and CE. However, due to the very small Params, the IT/Params, GE/Params, and CE/Params are higher compared to other methods. Besides, owing to the limited modeling ability of non-transformer architectures, VAE-WACGAN achieves lower Acc/F1 score compared to transformer-based methods. Specifically, AITrans improves the Acc/F1 score by 2.8%/2.6% and 2.4%/3.2% compared to VAE-WACGAN. But AITrans increases Params by 32.8 M, FLOPs by 28.9 G, GU by 15.3%, CU by 32.3%, GP by 38 W, CP by 22 W, IT by 8.2 ms, GE by 36.9E-8 kWh, CE by 19.4E-8 kWh. Among transformer-based methods, FlowTransformer [53] yields the largest FLOPs, which causes a high computational burden and a low inference speed. Compared to FlowTransformer, VAE-WACGAN reduces Params by 45.3 M, FLOPs by 68.9 G, GU by 20.0%, CU by 30.5%, GP by 43 W, CP by 23 W, IT by 9.6 ms, GE by 44.4E-8 kWh, CE by 22.8E-8 kWh. Due to the larger Params of FlowTransformer, its IT/Params, GE/Params, and CE/Params are lower than those of VAE-WACGAN. Compared to AITrans, FlowTransformer increases complexity in the main metrics. In summary, VAE-WACGAN is the fastest method with low complexity. By contrast, AITrans introduces higher complexity due to the transformer architecture, but the increase is acceptable. Compared to other transformer-based methods, AITrans still maintains relatively low complexity while achieving the best Acc/F1 score.

Compared to the basic transformer, the additional complexity of AITrans primarily comes from IHIA and ILIA. To further investigate the influence, this article analyzes the performance

TABLE III
ANALYSIS OF DIFFERENT WINDOW SEGMENTATION STRATEGIES

Window Size	IIoTA		IIoTB	
	Acc	F1 score	Acc	F1 score
Size = 10	96.6	96.2	97.5	96.6
Size = 20	97.0	96.9	98.0	97.0
Size = 50	97.6	97.4	98.6	97.8
Size = 100	97.3	97.2	98.1	97.2
Size = 200	96.8	96.7	97.7	97.0
AMSWs	99.1	98.7	99.6	99.2

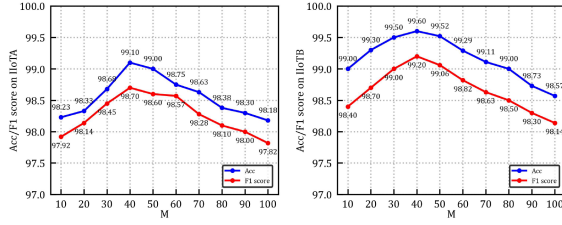
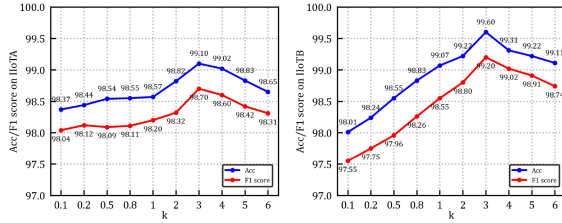
and complexity without IHIA/ILIA. The results show that embedding IHIA and ILIA brings significant performance improvements. Simultaneously, the parameters remain almost unchanged, since IHIA and ILIA perform computations on the existing attention without introducing many additional learnable parameters. Although the inclusion of IHIA and ILIA increases the model complexity to some extent, the overall complexity remains within an acceptable range.

D. Analysis of Training Convergence

To further analyze the training process of AITrans, the loss variation during the training process is illustrated in Fig. 4. On IIoTA, the loss starts from 7.3 and decreases rapidly during the early epochs. Afterwards, the rate of loss reduction gradually slows down. When the epoch reaches around 80, the loss gradually stabilizes at around 1.2. On IIoTB, the initial loss is 8.8, and a similar trend is observed with a rapid decline in the first stage, followed by a slower decrease until convergence at the 95th epoch, reaching a final loss of about 1.1. After convergence, the loss fluctuates slightly, but the amplitude of variation remains minimal, indicating the stability of the training process. These results confirm that AITrans exhibits efficient and stable convergence on IIoTA and IIoTB.

E. Analysis of AMSWS

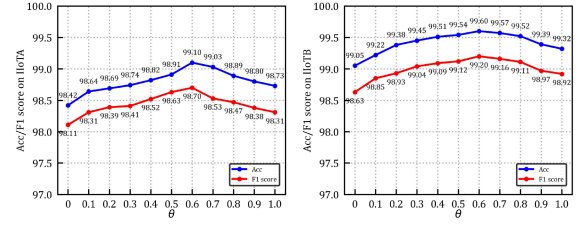
To validate the effectiveness of AMSWS, this article conducts comparative experiments against fixed-scale window segmentation (*Size* = 10/20/50/100/200) in Table III. The results demonstrate the effectiveness of AMSWS compared to fixed-scale window segmentation. On IIoTA and IIoTB, AMSWS consistently achieves superior performance, obtaining the highest Acc/F1 score. These results indicate that the adaptive window segmentation strategy can dynamically adjust the window to

Fig. 5. Analysis of parameter M .Fig. 6. Analysis of parameter k .

capture diverse temporal features of anomalies. Among fixed-scale methods, the optimal fixed window size is 50, yielding the Acc/F1 scores of 97.6%/97.4% on IIoTA and 98.6%/97.8% on IIoTB. Compared to this best-performing fixed-scale window, AMSWS enhances the Acc/F1 score by 1.5%/1.3% and 1.0%/1.4%. These improvements highlight the robustness of AMSWS to capture both short-term burst anomalies and long-term subtle anomalies in IIoT environments.

The parameter M in (5) controls the window size for traffic variation analysis. To investigate the impact of M on performance, comparative experiments are conducted with different M in Fig. 5. As M increases, the performance exhibits a rise-then-fall trend. Specifically, when M increases from 10 to 40, the model covers a longer temporal range, thereby enabling more comprehensive extraction of traffic variation characteristics. This leads to an improvement in performance, reaching the peak when $M = 40$. However, as M continues to increase, the excessively large window introduces redundant information and weakens the ability to capture critical traffic patterns. Moreover, an overly large window may smooth out short-term traffic fluctuations, thereby reducing the model's sensitivity to fine-grained anomalies. This causes a gradual decline in performance. These findings highlight that the choice of window size M is crucial for balancing feature coverage and effective information extraction. A window that is too small may fail to capture sufficient context, whereas a window that is too large may introduce redundancy and obscure anomalies. Therefore, this parameter is set to $M = 40$.

The parameters k and θ in (6) determine the window sensitivity to traffic variation. To validate the effectiveness and analyze the effect of k and θ on performance, comparative experiments are conducted in Figs. 6 and 7. As k and θ increase, the performance first rises and then declines. This confirms that the moderate increase allows the model to capture traffic fluctuations more effectively. When k is too large, the model becomes overly sensitive to traffic fluctuations, which tends to amplify noise or non-critical variations. When θ is too large, many fine-grained

Fig. 7. Analysis of parameter θ .TABLE IV
ANALYSIS OF IHIA AND ILIA

Attention		IIoTA		IIoTB	
IHIA	ILIA	Acc	F1 score	Acc	F1 score
×	×	97.2	96.4	98.1	97.3
✓	×	98.3	97.9	99.0	98.5
×	✓	98.0	97.3	98.6	98.1
✓	✓	99.1	98.7	99.6	99.2

anomalies are smoothed out and ignored, which causes the model to lose sensitivity to critical attacks. The best performance is achieved when $k = 3$ and $\theta = 0.6$. In contrast, without introducing these parameters ($k = 1$ and $\theta = 0$), the performance drops. Specifically, when $k = 1$, the Acc/F1 score decreases to 98.57%/98.20% on IIoTA and 99.07%/98.55% on IIoTB. When $\theta = 0$, the Acc/F1 score decreases to 98.42%/98.11% on IIoTA and 99.05%/98.63% on IIoTB. These results clearly demonstrate the effectiveness of introducing k and θ . By controlling the sensitivity of the window to traffic variations, the model can effectively enhance the ability to represent critical anomalies. Therefore, these parameters are set to $k = 3$ and $\theta = 0.6$.

F. Analysis of AITrans

In AITrans, AHI includes IHIA and ILIA. To validate the effectiveness of IHIA and ILIA, this article conducts ablation studies in Table IV. Experimental results show that using either IHIA or ILIA provides noticeable performance improvements compared to the standard transformer. Specifically, using single IHIA brings the Acc/F1 score improvements of 1.1%/1.5% on IIoTA and 0.9%/1.2% on IIoTB, while applying single ILIA brings enhancements of 0.8%/0.9% and 0.5%/0.8%. It indicates that both modules individually strengthen the capability to model complex anomaly patterns. Furthermore, when IHIA and ILIA are combined, the model achieves optimal performance. This highlights the complementary nature of the two proposed attention mechanisms. The inter-head and inter-layer attention interaction collaboratively enhance the transformer's ability to identify anomalous IIoT traffic patterns.

To further evaluate the impact of IHIA and ILIA on performance, this article compares the intra-class feature distances and inter-class feature distances under different attention mechanisms in Fig. 8. The results show that on IIoTA and IIoTB, in the absence of any attention enhancement, intra-class feature distances remain relatively large, while inter-class distances are comparatively small. There is a significant overlap between distance distributions. Introducing either IHIA or ILIA can effectively reduce intra-class distances and increase inter-class distances, thereby narrowing the overlap region and improving

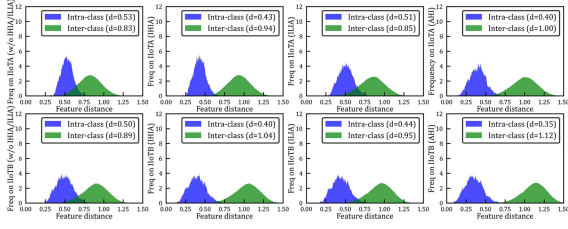


Fig. 8. Analysis of intraclass and interclass feature distance.

 TABLE V
ANALYSIS OF ATTENTION AGGREGATION FROM PREVIOUS LAYERS

Attention Aggregation	IIoTA		IIoTB	
	Acc	F1 score	Acc	F1 score
Weighted Aggregation	98.7	98.2	99.2	99.0
Previous $P = 1$ Layer	98.1	97.5	98.3	97.8
Previous $P = 2$ Layer	98.3	97.6	98.4	98.3
Previous $P = 3$ Layer	98.5	97.9	98.8	98.7
Previous $P = 4$ Layer	98.7	98.3	99.1	98.8
Previous $P = 5$ Layer	98.8	98.4	99.3	98.9
Previous $P = 6$ Layer	98.9	98.6	99.5	99.2
Previous $P = 7$ Layer	99.1	98.7	99.6	99.2

the inter-class separability of different classes. Notably, individual IHIA yields better results than ILIA, which aligns with the findings from quantitative analysis. When both IHIA and ILIA are integrated, the model achieves the most significant performance improvement, with efficient optimization of both intra-class compactness and inter-class separability. These experimental findings confirm the effectiveness of the proposed attention enhancement strategies.

ILIA integrates attention from all previous transformer layers. To validate the effectiveness, the performance when using the attention from the previous P layers is analyzed. AITrans consists of eight layers, $P = 1/2/3/4/5/6/7$ can be set. $P = 7$ means aggregating the attention from all previous layers. In addition, this article compares the performance under a weighted aggregation strategy, in which the weights for layers gradually increase as the layer deepens, the results are in Table V. With the increase in aggregated layers, the performance improves. This means aggregating information from more layers can capture richer features. Compared to using only the attention from the adjacent layer ($P = 1$), where the Acc/F1 score of 98.1%/97.5% on IIoTA and 98.3%/97.8% on IIoTB are achieved, the performance when aggregating all previous layers ($P = 7$) is improved by 1.0%/1.2% and 1.3%/1.4%. The weighted aggregation strategy also achieves relatively high performance, but it is slightly inferior to the best result at $P = 7$. This indicates that although weighted aggregation also leverages information from all layers, the gradually increasing weights weaken the influence of shallow information. This causes fine-grained patterns not to be fully utilized. Simultaneously, the excessive emphasis on deeper layers causes the model to overly rely on high-level features.

This article further analyzes the parameters α and β in AHI as demonstrated in Figs. 9 and 10. As α increases from 0 to 1, the performance on IIoTA and IIoTB initially improves and then declines, achieving optimal results at $\alpha = 0.3$. Similarly, as β increases from 0 to 1, the performance also exhibits an initial upward and subsequent downward trend, with the best

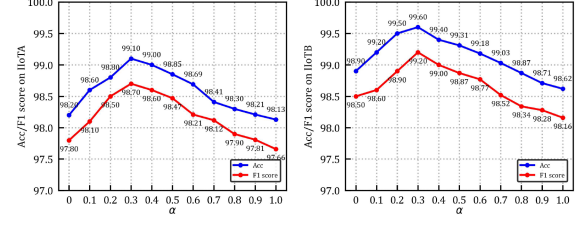
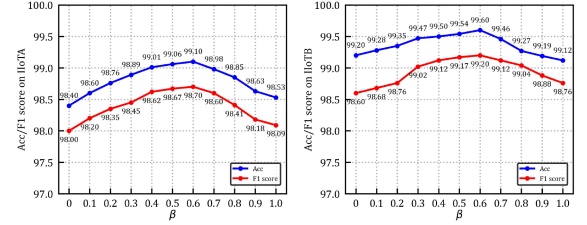
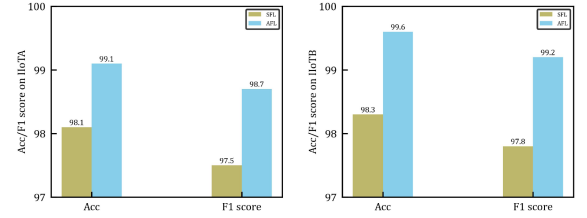

 Fig. 9. Analysis of parameter α .

 Fig. 10. Analysis of parameter β .


Fig. 11. Analysis of SFL and AFL.

 TABLE VI
ANALYSIS OF ACW AND HAA

Optimization	IIoTA		IIoTB	
	ACW	HAA	Acc	F1 score
$\times$	$\times$	$\times$	98.1	97.5
$\checkmark$	$\times$	$\times$	98.8	98.4
$\times$	$\checkmark$	$\times$	98.6	98.3
$\checkmark$	$\checkmark$	$\checkmark$	99.1	98.7
			98.3	97.8
			98.9	98.7
			99.0	98.4
			99.6	99.2

performance observed when $\beta = 0.6$. In AHI, such two parameters can balance the original information and the aggregated information. When an appropriate balance is properly maintained during information fusion, the interaction mechanism can maximally enhance feature representations. Therefore, these parameters are set to $\alpha = 0.3$ and $\beta = 0.6$.

G. Analysis of AFL

To validate the effectiveness of AFL, experiments are conducted against the standard focal loss (SFL). The results in Fig. 11 indicate that AFL outperforms SFL. AFL improves the Acc/F1 score by 1.0%/1.2% and 1.3%/1.4%. These results demonstrate that AFL effectively alleviates the issue of class imbalance. It enables the model to better detect rare intrusions that are difficult to identify. To investigate the contributions of ACW and HAA in AFL, this article conducts ablation experiments in Table VI. The results show that both ACW and HAA enhance the performance. Specifically, using individual ACW brings the Acc/F1 score improvements of 0.7%/0.9% on IIoTA and

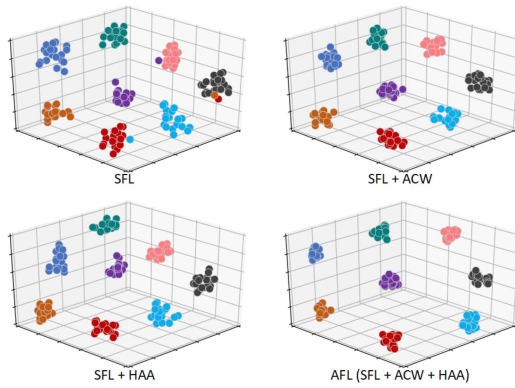


Fig. 12. Feature distribution under different losses.

0.6%/0.9% on IIoT. Employing individual HAA also brings the Acc/F1 score improvements of 0.5%/0.8% and 0.7%/0.6%. Moreover, when both ACW and HAA are combined to optimize the focal loss, the model achieves optimal performance. This indicates that these two mechanisms complement each other and jointly enhance the intrusion detection capability for rare samples in IIoT environments.

To further investigate the advantages brought by ACW and HAA, the feature distributions are visualized in Fig. 12. When using SFL, the distribution exhibits relatively large intra-class distances and noticeable outliers within the same class. After incorporating either ACW or HAA optimization, intra-class distances become more compact, and the number of outliers significantly decreases. Moreover, when both ACW and HAA are employed simultaneously, the distribution demonstrates further reduction of intra-class distances and fewer outliers. This forms more distinct and compact clusters, and the samples with the same label are aggregated more closely together. This indicates that the proposed AFL can effectively optimize the feature representation. It leads to enhanced intrusion detection performance and improves the robustness.

V. CONCLUSION

This article proposed an IIoT intrusion detection approach. First, an AMSWS strategy was proposed to dynamically adjust the segmentation window via traffic variation. This accurately captured anomalies exhibiting diverse temporal characteristics, thus effectively detecting various types of intrusion patterns. Second, an AITrans comprising IHIA and ILIA was proposed to facilitate effective inter-head and inter-layer attention interaction. This enhanced the capability for representing complex anomaly patterns. Finally, an AFL comprising ACW and HAA was proposed to mitigate the class imbalance issue. This enabled the model to better detect minority-class anomalous samples.

Despite the competitive performance achieved by the proposed method in IIoT intrusion detection, several limitations remain. First, since AITrans is built upon the transformer, the complexity is relatively higher compared to non-transformer approaches. This may limit the scalability when deployed in extremely large-scale IIoT networks with limited resources.

Second, although AFL can mitigate class imbalance, there remains an overfitting risk to the majority-class samples in highly imbalanced datasets. This may lead to reduced sensitivity to rare and stealthy intrusion types. Future research will investigate lightweight transformer variants, compression strategies, and distributed architectures to improve the scalability of AITrans for deployment in large-scale IIoT networks. In addition, more advanced imbalance-aware learning strategies and data augmentation technologies will be explored to further enhance robustness under highly imbalanced scenarios.

REFERENCES

- [1] P. Ruzafa-Alcázar et al., "Intrusion detection based on privacy-preserving federated learning for the industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1145–1154, Feb. 2023.
- [2] J. Pei et al., "Neuro-vae-symbolic dynamic traffic management," *IEEE Trans. Intell. Transp. Syst.*, to be published, doi: [10.1109/TITS.2025.3571210](https://doi.org/10.1109/TITS.2025.3571210).
- [3] J. Zhang, C. Luo, M. Carpenter, and G. Min, "Federated learning for distributed IIoT intrusion detection using transfer approaches," *IEEE Trans. Ind. Informat.*, vol. 19, no. 7, pp. 8159–8169, Jul. 2023.
- [4] J. Pei, M. Omar, M. M. Al Dabel, S. Mumtaz, and W. Liu, "Federated few-shot learning with intelligent transportation cross-regional adaptation," *IEEE Trans. Intell. Transp. Syst.*, to be published, doi: [10.1109/TITS.2025.3563928](https://doi.org/10.1109/TITS.2025.3563928).
- [5] V. Sobchuk, R. Pykhivskyi, O. Barabash, S. Korotin, and S. Omarov, "Sequential intrusion detection system for zero-trust cyber defense of iot/iiot networks," *Adv. Inf. Syst.*, vol. 8, no. 3, pp. 92–99, 2024.
- [6] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "Ton_iot telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.
- [7] T. Wang, K. Fang, W. Wei, J. Tian, Y. Pan, and J. Li, "Microcontroller unit chip temperature fingerprint informed machine learning for iiot intrusion detection," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 2219–2227, Feb. 2023.
- [8] C. R. R. PR and K. Baalaji, "A graphics processing unit assisted CNN-GRU framework for the intrusion detection mechanism in the industrial Internet of Things," *Eng. Res. Express*, vol. 7, 2025, Art. no. 025240.
- [9] M. Arsalan, M. Mubeen, M. Bilal, and S. F. Abbasi, "1d-cnn-ids: 1dcnn-based intrusion detection system for iiot," in *Proc. 29th Int. Conf. Autom. Comput.*, 2024, pp. 1–4.
- [10] W. Mei, W. Liu, J. Chen, and K. Li, "A physical signal-based anomaly detection for industrial terminal," in *Proc. 7th Int. Conf. Electron. Inf. Technol. Comput. Eng.*, 2023, pp. 651–658.
- [11] M. Tayebi and S. El Kadhali, "Performance analysis of recurrent neural networks for intrusion detection systems in industrial-Internet of Things," *Franklin Open*, vol. 12, 2025, Art. no. 100310.
- [12] K. Bansal and A. Singhrova, "Review on intrusion detection system for IOT/IIOT-brief study," *Multimedia Tools Appl.*, vol. 83, no. 8, pp. 23083–23108, 2024.
- [13] M. O. Alassafi, S. H. Hasan, S. Badri, and S. H. Hasan, "Optimized bi-LSTM: A novel approach for attack detection in industrial IoT," *Signal, Image Video Process.*, vol. 18, no. 5, pp. 4903–4913, 2024.
- [14] Z. Chen, Z. Li, J. Huang, S. Liu, and H. Long, "An effective method for anomaly detection in Industrial Internet of Things using XGBoost and LSTM," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 23969.
- [15] M. Wu, Y. Zheng, D. S.-H. Wong, Y. Wang, and X. Hu, "Tracer: Attack-aware divide-and-conquer transformer for intrusion detection in industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 21, no. 6, pp. 4924–4934, Jun. 2025.
- [16] J. Casajús-Setién, C. Bielza, and P. Larrañaga, "Anomaly-based intrusion detection in IIoT networks using transformer models," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience*, 2023, pp. 72–77.
- [17] H. Gao, W. Jiang, Q. Ran, K. Wang, X. Ma, and Y. Xu, "Srt: A skip-range transformer for detecting anomalies in multi-attribute industrial time series data," *IEEE Internet Things J.*, vol. 12, no. 19, pp. 39223–39236, Oct. 2025.
- [18] Z. Zamanzadeh Darban, G. I. Webb, S. Pan, C. Aggarwal, and M. Salehi, "Deep learning for time series anomaly detection: A survey," *ACM Comput. Surv.*, vol. 57, no. 1, pp. 1–42, 2024.

- [19] Z. Wang, Y. Wang, C. Gao, F. Wang, T. Lin, and Y. Chen, "An adaptive sliding window for anomaly detection of time series in wireless sensor networks," *Wireless Netw.*, vol. 28, no. 1, pp. 393–411, 2022.
- [20] S. Sun, M. Gu, and T. Liu, "Adaptive sliding window–dynamic time warping–based fluctuation series prediction for the capacity of lithium-ion batteries," *Electronics*, vol. 13, no. 13, 2024, Art. no. 2501.
- [21] G. Papageorgiou and C. Tjortjij, "Adaptive sliding window normalization," *Inf. Syst.*, vol. 129, 2025, Art. no. 102515.
- [22] K. Chang and S. H. Park, "Reinforcement learning-based real-time contrasts control chart using an adaptive window size," *IEEE Access*, vol. 13, pp. 72077–72086, 2025.
- [23] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 10012–10022.
- [24] S. Luo, Z. Zhao, Q. Hu, and Y. Liu, "A hierarchical CNN-transformer model for network intrusion detection," in *Proc. 2nd Int. Conf. Appl. Math., Model., Intell. Comput.*, 2022, vol. 12259, pp. 853–860.
- [25] P. Liang, L. Yang, Z. Xiong, X. Zhang, and G. Liu, "Multilevel intrusion detection based on transformer and wavelet transform for IoT data security," *IEEE Internet Things J.*, vol. 11, no. 15, pp. 25613–25624, Aug. 2024.
- [26] T. Jiang, X. Fu, and M. Wang, "BBO-CFAT: Network intrusion detection model based on BBO algorithm and hierarchical transformer," *IEEE Access*, vol. 12, pp. 54191–54201, 2024.
- [27] H. Mancy and Q. H. Naith, "Swinot: A hierarchical transformer-based framework for behavioral anomaly detection in IoT-driven smart cities," *IEEE Access*, vol. 13, pp. 48758–48774, 2025.
- [28] M. A. Qathradly et al., "Saccn-ids: A self-attention convolutional neural network for intrusion detection in Industrial Internet of Things," *CAAI Trans. Intell. Technol.*, vol. 9, no. 6, pp. 1398–1411, 2024.
- [29] S. Li et al., "Crfsf: An intrusion detection framework for Industrial Internet of Things based on pretrained CNN2D-RNN and SVM," *IEEE Access*, vol. 11, pp. 92041–92054, 2023.
- [30] G. Zeng, Y. Yang, K. Lu, G. Geng, and J. Weng, "Evolutionary adversarial autoencoder for unsupervised anomaly detection of Industrial Internet of Things," *IEEE Trans. Rel.*, vol. 74, no. 3, pp. 3454–3468, Sep. 2025.
- [31] K. Lu, J. Huang, G. Zeng, M. Chen, G. Geng, and J. Weng, "Multi-objective discrete extremal optimization of variable-length blocks-based CNN by joint NAS and HPO for intrusion detection in IIoT," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 4, pp. 4266–4283, Jul./Aug. 2025.
- [32] S. E. Vadakkethil, K. Polimetla, Z. Alsalmi, P. K. Pareek, and D. Kumar, "Mayfly optimization algorithm with bidirectional long-short term memory for intrusion detection system in Internet of Things," in *Proc. 3rd Int. Conf. Distrib. Comput. Elect. Circuits Electron.*, 2024, pp. 1–4.
- [33] K. Yang, J. Wang, and M. Li, "An improved intrusion detection method for IIoT using attention mechanisms, bigru, and inception-CNN," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 19339.
- [34] I. A. Fares, M. Abd Elaziz, A. O. Aseeri, H. S. Zied, and A. G. Abdellatif, "Tfkan: Transformer based on Kolmogorov-Arnold networks for intrusion detection in IoT environment," *Egyptian Informat. J.*, vol. 30, 2025, Art. no. 100666.
- [35] P. S. Moghaddam, A. Vaziri, S. S. Khatami, F. Hernando-Gallego, and D. Martín, "Generative adversarial and transformer network synergy for robust intrusion detection in IoT environments," *Future Internet*, vol. 17, no. 6, pp. 258–293, 2025.
- [36] P. Wang, Y. Song, X. Wang, and X. Guo, "Diff: A diffusion-transformer for intrusion detection of IoT with imbalanced learning," *J. Netw. Syst. Manage.*, vol. 33, no. 3, pp. 48–71, 2025.
- [37] Z. Zeng, X. Deng, B. Zhao, X. Liu, J. Zheng, and J. Chen, "Toward intelligent attack detection with causal transformer in Internet of Things," *IEEE Internet Things J.*, vol. 12, no. 8, pp. 10751–10767, Apr. 2025.
- [38] P. Wang, Y. Song, X. Wang, X. Guo, and Q. Xiang, "Imagids: An Internet of Things intrusion detection framework utilizing GADF imaging encoding and improved transformer," *Complex Intell. Syst.*, vol. 11, no. 1, pp. 93–108, 2025.
- [39] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.
- [40] Q. Liu, S. Lee, and J. Paparrizos, "TSB-autoad: Towards automated solutions for time-series anomaly detection," *Proc. VLDB Endowment*, vol. 18, no. 11, pp. 4364–4379, 2025.
- [41] Z. Liu, H. Xu, Y. Kuang, and F. Li, "Svmdformer: A semi-supervised vehicular misbehavior detection framework based on transformer in IOV," in *Proc. IEEE 43rd Int. Conf. Distrib. Comput. Syst.*, 2023, pp. 887–897.
- [42] R. B. Basnet, R. Shash, C. Johnson, L. Walgren, and T. Doleck, "Towards detecting and classifying network intrusion traffic using deep learning frameworks," *J. Internet Serv. Inf. Secur.*, vol. 9, no. 4, pp. 1–17, 2019.
- [43] P. D. Kingma and L. J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Representations*, 2015, pp. 1–15.
- [44] N. Alkhatib, M. Mushtaq, H. Ghauch, and J.-L. Danger, "Can-bert do it? controller area network intrusion detection system based on BERT language model," in *Proc. IEEE/ACS 19th Int. Conf. Comput. Syst. Appl.*, 2022, pp. 1–8.
- [45] H. Hayouni and L. Nasraoui, "NAIDS4IIoT: A novel artificial intelligence-based intrusion detection architecture for the Internet of Things," *Intelligence Artif.*, vol. 28, no. 76, pp. 253–282, 2025.
- [46] A. Ericson, S. Forsström, and K. Thar, "IIoT intrusion detection using lightweight deep learning models on edge devices," in *Proc. IEEE 20th Int. Conf. Factory Commun. Syst.*, 2024, pp. 1–8.
- [47] S. Latif et al., "Intrusion detection framework for the Internet of Things using a dense random neural network," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 6435–6444, Sep. 2022.
- [48] Z. E. Huma et al., "ACNN-IDS: An attention-based CNN for cyberattack detection in IoT," in *Proc. 2nd Int. Conf. Cyber Resilience*, 2024, pp. 1–6.
- [49] A. Thaljaoui, "Intelligent network intrusion detection system using optimized deep CNN-LSTM with uns-wb15," in *Proc. Int. J. Inf. Technol.*, 2025, pp. 1–17.
- [50] Y. A. Rani and E. S. Reddy, "Deep intrusion net: An efficient framework for network intrusion detection using hybrid deep TCN and GRU with integral features," *Wireless Networks*, vol. 31, no. 2, pp. 1255–1278, 2025.
- [51] W. Tian, Y. Shen, N. Guo, J. Yuan, and Y. Yang, "VAE-WACGAN: An improved data augmentation method based on VAEGAN for intrusion detection," *Sensors (Basel, Switzerland)*, vol. 24, no. 18, 2024, Art. no. 6035.
- [52] Z. Ding et al., "MF-Net: Multi-frequency intrusion detection network for internet traffic data," *Pattern Recognit.*, vol. 146, 2024, Art. no. 109999.
- [53] L. D. Manocchio, S. Layeghy, W. W. Lo, G. K. Kulatilake, M. Sarhan, and M. Portmann, "Flowtransformer: A transformer framework for flow-based network intrusion detection systems," *Expert Syst. Appl.*, vol. 241, 2024, Art. no. 122564.



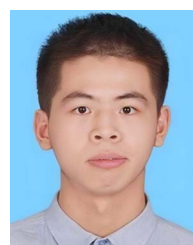
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